Example Accompaniment

Lecture 2: Formulating research question

ICMR INDIAN COUNCIL OF MEDICAL RESEARCH | NIE NATIONAL INSTITUTE OF EPIDEMIOLOGY

Basic Course in Biomedical Research

As mandated by the National Medical Commission

Lecture 2: Formulating a research question

Objective

To demonstrate how to write research question, PICO statement, study objective and study title using examples.

Background

After finalising the topic in the previous section, the PG should now frame the Research Question, the PICO statement *(PICO is an acronym for Population, Intervention, Comparison, Outcome. Sometimes it is referred to as 'PICOT' with T to denote Time)*, the study objectives and the study title.

Refining the research question and formulation of PICO

In the previous section, we saw that the PG finalised the query, "What is the percentage of urinary incontinence (existing and newly developed) among women with third degree uterine prolapse who had undergone vaginal hysterectomy in her hospital?"

Now the research question has to be refined in the following manner.

What is the prevalence of urinary incontinence among adult women who underwent vaginal hysterectomy for third degree uterine prolapse in a tertiary care hospital?

This research question can be examined using a descriptive cross-sectional study (see Lecture 5). This is one of the most commonly used basic study designs in epidemiology and clinical research. In this design, eligible participants will be interviewed only once (after 3 months of surgery but before completing 4 months) to assess the presence/absence of urinary incontinence (existing and newly developed). The PICO statement for a descriptive cross-sectional study does not have 'I' and 'C'. So, the PICO can be written as follows:

Population	Adult women (aged ≥18 years) who underwent vaginal hysterectomy for third degree uterine prolapse in a tertiary care hospital
Intervention	Not applicable
Comparison	Not applicable
Outcome	Prevalence of urinary incontinence (existing and newly developed) within 3 months after hysterectomy
Time	XXX-XXX 2020

Note: The PICO statement is usually written for the primary objective of a study.

Framing the study objectives

Following the guidelines of framing study objectives as described in the lecture, the PG writes:

Primary objective

1. To estimate the prevalence of urinary incontinence (within 3 months of surgery) among adult women who underwent vaginal hysterectomy for third degree uterine prolapse in a tertiary care hospital during XXX-XXX 2020.

Secondary objective

2. To determine the demographic and clinical factors associated with urinary incontinence among adult women who underwent vaginal hysterectomy for third degree uterine prolapse in a tertiary care hospital during XXX-XXX 2020.

Framing the study title

The PG writes the following as study title, keeping in mind that the title should be precise, concise and informative (time, place, person, exposure, outcome and study design).

Prevalence and determinants of urinary incontinence after vaginal hysterectomy for uterine prolapse in a tertiary care hospital during XXX-XXX 2020 – a cross-sectional study

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Variations of the RQ and their PICO statement

Now, we will look at certain other types of research question and see how to construct the corresponding PICO statements. The more description you put in your research question the better it is.

Scenario 1: *What are the determinants of urinary incontinence among the adult women with third degree uterine prolapse who have undergone hysterectomy in a tertiary care facility?*

This research question can be answered using an analytical cross-sectional study design or a case-control design.(Lecture 6) In this case, the 'I' in PICO will stand for the presence of risk factor or exposure (e.g., Smoker) and 'C' will stand for the counterfactual i.e., absence of risk factor or exposure (e.g., Non-smoker). Let's assume that we want to examine 'age group' (older vs. younger women) as a determinant for urinary incontinence in this study population and the PICO can be written as follows.

Population	Adult women (age ≥18 years) diagnosed with third degree uterine prolapse who have undergone hysterectomy in a tertiary care facility
Risk factor/ exposure	Older women (>60 years)
Comparison	Younger women (18-60 years)
Outcome	Urinary incontinence
Time	XXX-XXX 2020

You may have many more risk factors/exposures to be examined in your study. For instance, in addition to age, you may wish to examine the effect of tobacco intake, diabetes, hypertension, duration of illness, type of surgeon and socio-economic status in the above example. Here, you can write your I and C in a generic manner.

A case-control design (Lecture 6) for this research question would start with recruiting adult women who have undergone hysterectomy for third degree uterine prolapse. These women will be divided into cases (those with urinary incontinence) and controls (those without urinary incontinence). Then, all the exposures of interest will be measured and compared between the cases and controls.

A cohort study design is also possible for the same PICO. A cohort of women who underwent hysterectomy and do not have urinary incontinence will be recruited. At baseline, details about the presence/absence of exposures will be recorded. Those with the exposure will be labelled 'exposed' (for e.g., age >60 years) and those without the exposure will be labelled 'unexposed' (18-60 years). The cohort will be then followed up for say, 6 months, to record the development of new urinary incontinence. Those who developed urinary incontinence will be considered as diseased and those who do not develop it will be considered as non-diseased. Then, the diseased and non-diseased groups will be compared to identify the risk factors.

Scenario 2: *What is the difference in the level of incidence of urinary incontinence between laparoscopic hysterectomy and open abdominal hysterectomy performed for uterine prolapse among adult women in a tertiary care hospital?*

This research question can be best answered using a Randomised Controlled Trial (Lecture 7) in which women with uterine prolapse in need of surgery could be randomised into two groups – one receiving the laparoscopic type and the other receiving an open abdominal type. The two groups will be followed for say, 6 months, to measure and compare the incidence of urinary incontinence (newly developed).

Population	Adult women (aged ≥18 years) diagnosed with uterine prolapse who require hysterectomy in a tertiary hospital
Intervention	Laparoscopic hysterectomy
Comparison	Open abdominal hysterectomy
Outcomes	Urinary incontinence
Time	XXX-XXX 2020

The above question can be answered through observational studies also. A prospective cohort study is possible here (Lecture 6). This would begin with recruiting women undergoing any of the two types of surgery and they would be followed up for say, 6 months, to measure and compare the incidence of urinary incontinence between the two groups. A retrospective cohort study using medical records is also possible here. Such a study will begin with retrieving the hospital records of patients who underwent any one of the two types of surgery for uterine prolapse during a specified period. Subsequently, the records can be examined for follow-up visits where the complaint 'urinary incontinence' is documented. However, record-based studies are often problematic because they do not always have the necessary information recorded uniformly and reliably. In that case, the records can help in recruiting the cohort initially and the outcomes can be measured prospectively.

Concluding remarks

It is very important to spend a considerable amount of time in refining your research question, and framing your study objectives. This is because all other subsequent steps in research are dependent on them. The choice of a study design should be based on the research objective and practicality.

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